

Dependency Preserving Decomposition

Normalization Properties • Projection of FDs • Equivalence Check • 3NF vs BCNF

When decomposing a database relation R into smaller relations $(R_1, R_2, \dots, R_n)$ to eliminate redundancy, we must ensure that all functional dependencies (FDs) present in the original schema can still be enforced locally across the decomposed tables without needing expensive inter-table joins.

★ Formal Definition

A decomposition of relation R with FD set F into $R_1, R_2, \dots, R_n$ is **Dependency Preserving** if and only if the union of the projected dependency sets on each sub-relation is equivalent to the original set F :

$$F \equiv F' \quad \text{where} \quad F' = F_1 \cup F_2 \cup \dots \cup F_n$$

1. The 5-Step Exam Verification Procedure

Follow this exact methodology when solving university exam or GATE questions:



💡 Pro-Tip for Fast Exam Solving

Since projected dependencies F_i are subsets/derivations of F , **F covers F'** is automatically true! Therefore, in practice, you only need to verify if **F' covers F** (i.e., can every original FD in F be derived using F' ?).

2. Solved Examples

Example 1: Dependency Preserving Decomposition

Given relation $R(A, B, C, D)$ with original FDs:

$$F = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$$

Decomposed into: $R_1(A, B)$, $R_2(B, C)$, and $R_3(C, D)$.

Analysis

- Projected FDs for $R_1(A, B)$: $F_1 = \{ A \rightarrow B \}$
- Projected FDs for $R_2(B, C)$: $F_2 = \{ B \rightarrow C \}$
- Projected FDs for $R_3(C, D)$: $F_3 = \{ C \rightarrow D \}$

Combined Projected Set: $F' = F_1 \cup F_2 \cup F_3 = \{ A \rightarrow B, B \rightarrow C, C \rightarrow D \}$

Since F' is identical to F , $F \equiv F'$ holds trivially.

Result: Decomposition is Dependency Preserving ✓

Example 2: GATE-Style Non-Preserving Decomposition

Given relation $R(A, B, C, D)$ with original FDs:

$F = \{ A \rightarrow B, C \rightarrow D, B \rightarrow C \}$

Decomposed into: $R_1(A, B)$, $R_2(B, C)$, and $R_3(B, D)$.

Step 1: Identify Projected FDs

- In $R_1(A, B)$: FD $A \rightarrow B$ holds directly.
- In $R_2(B, C)$: FD $B \rightarrow C$ holds directly.
- In $R_3(B, D)$: Can we enforce $B \rightarrow D$? From F , $B^+ = \{B, C, D\}$, so $B \rightarrow D$ holds on R_3 .

Combined Projected Set: $F' = \{ A \rightarrow B, B \rightarrow C, B \rightarrow D \}$

Step 2: Check if F' covers original F

We test each FD in F using attribute closure under F' :

- Test $A \rightarrow B$: Present directly in F' ✓ **Preserved**
- Test $B \rightarrow C$: Present directly in F' ✓ **Preserved**
- Test $C \rightarrow D$: Calculate C^+ using $F' = \{ A \rightarrow B, B \rightarrow C, B \rightarrow D \}$:
 - Start: $C^+ = \{C\}$
 - Can we apply $A \rightarrow B$? ✗ No A .
 - Can we apply $B \rightarrow C$ or $B \rightarrow D$? ✗ No B .
 - Final $C^+ = \{C\}$.

Since $D \notin C^+ \rightarrow$ ✗ **Failed! $C \rightarrow D$ cannot be enforced in any decomposed relation.**

Result: Decomposition is NOT Dependency Preserving ✗

3. Dependency Preservation vs. Lossless Join

These are two independent properties of database decomposition. Do not confuse them!

Feature	Lossless Join Decomposition	Dependency Preserving Decomposition
Primary Goal	Ensures data integrity & prevents spurious tuples	Ensures constraints (FDs) are easily enforceable

Mechanism	Natural join of decomposed tables yields original table	Union of projected FDs covers original FD set
Condition	$R_1 \cap R_2 \rightarrow R_1$ or $R_1 \cap R_2 \rightarrow R_2$	$F \equiv F_1 \cup F_2 \cup \dots \cup F_n$
Impact if Lost	Incorrect query results (extra false rows)	Inter-table joins required to check updates/inserts

4. Normal Forms Trade-off (3NF vs. BCNF)

3NF Synthesis Algorithm

Guarantees **BOTH** properties simultaneously:  **Always Lossless Join** AND  **Always Dependency Preserving**.

BCNF Decomposition

Guarantees  **Always Lossless Join**, BUT  **Dependency Preservation is NOT guaranteed**. In BCNF, we sometimes have to sacrifice dependency preservation to achieve complete redundancy elimination.

Quick Decision Formula:

Collected FDs (F') $\equiv$ Original FDs (F) $\Rightarrow$ Dependency Preserved 

One-Line Exam Summary

“Decompose the table, collect projected FDs, take their union, and test whether all original FDs can still be derived using attribute closure.”